16 Drugs and Supplements in Athletic Injury Recovery and Performance: WADA Perspective

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Abstract

The use of drugs and nutritional supplements to enhance sports and exercise performance has been widespread at all levels of participation and competition, albeit to a varying extent. The World Anti-doping Agency (WADA) was established to address the problem of drug use by athletes for their potential ergogenic effects. The primary purpose of WADA is to protect athletes worldwide, promote the values of clean sport, and preserve the spirit of sport. WADA's primary role is to develop, harmonize, and coordinate anti-doping rules and policies across all sports and countries. In addition, WADA engages in a wider range of other activities including scientific and social science research; education; intelligence and investigations; development of anti-doping capacity; and monitoring of compliance with the World Anti-Doping Program. The detection of a prohibited substance or its metabolite or laboratory markers in an athlete's sample (urine, blood, or other as appropriate) constitutes doping. The categories of WADA-prohibited substances and methods include anabolic agents, peptide hormones and their releasing factors, growth factors and growth modulators, erythropoiesis promoting agents and methods, beta-2 agonists, hormone and metabolic modulators, and masking agents such as diuretics. Optimal baseline nutrition, treating any micronutrient deficiencies, and maintaining appropriate fluid and electrolyte balance are all essential for optimal sports performance. Although a large number of nutritional supplements are being marketed for their purported ergogenic effects, only a handful have been shown to have such effect to a variable degree; these include caffeine, creatine, nitrate, beta-alanine, and sodium bicarbonate.

Keywords

World Anti-doping Agency · Prohibited substances · Prohibited methods · Athlete Biological Passport · Caffeine · Creatine

Introduction

The World Anti-Doping Agency (WADA) is a not-for-profit foundation that was set up under the initiative of the International Olympic Committee (IOC), which is supported by intergovernmental organizations, governments, public authorities, and other public and private bodies fighting doping in sport (WADA 2023a). WADA's primary role is to develop, harmonize, and coordinate anti-doping rules and policies across all sports and countries. In addition, WADA engages in a wide range of other activities including scientific and social science research; education; intelligence and investigations;

development of anti-doping capacity; and monitoring of compliance with the World Anti-Doping Program (WADA 2023a, b).

The WADA, through its anti-doping program, sets forth specific anti-doping rules and principles that are to be followed by organizations responsible for adopting, implementing, or enforcing anti-doping rules within their authority (WADA 2023b). Anti-doping rules, like competition rules, are sport rules governing the conditions under which sport is played.

The key aims of WADA, operationalized via its anti-doping program, are to protect the athlete's right to participate in doping-free sport, promote fairness and equity in sport participation, and ensure implementation of harmonized, coordinated, and effective preventive anti-doping programs worldwide (WADA 2023a, b). The detection of a prohibited substance or its metabolite or laboratory markers in an athlete's sample (urine, blood, or other as appropriate) constitutes doping (WADA 2023a, b). WADA has stipulated several other elements, violation of any one or more of which also constitutes doping; these address processes and procedures, protocols, and compliance issues integral to anti-doping activities (WADA 2023a, b).

Prohibited Substances and Prohibited Methods

The WADA anti-doping program applies the following key criteria for including substances and methods on the prohibited list: (1) medical or other scientific evidence, pharmacological effect or experience that the substance or method, alone or in combination with other substances or methods, has the potential to enhance or enhances sport performance; represents an actual or potential health risk to the athlete; has the potential to mask the use of other prohibited substances or methods, and (2) WADA's determination that the use of the substance or method violates the spirit of sport described in its anti-doping program (WADA 2023a, b).

According to WADA, the presence of a prohibited substance or its metabolites or markers, and/or the use or attempted use, possession or administration or attempted administration of a prohibited substance or prohibited method shall not be considered an anti-doping rule violation if it is consistent with the provisions of a therapeutic use exemption standard (WADA 2023b, c). A therapeutic use exemption (TUE) allows an athlete with a medical condition to use a prohibited substance or prohibited method, under certain specified conditions (WADA 2023c).

Substances That Are Prohibited at All Times

Non-approved Substances

Any pharmacological substance that is not included in the prohibited list and with no current approval by any governmental regulatory health authority for human therapeutic use (e.g., drugs under pre-clinical or clinical development or discontinued, designer drugs, substances approved only for veterinary use) is prohibited at all times (WADA 2023c).

Anabolic androgenic steroids and other anabolic agents (AASs) (Choong et al. 2011; Chrousos 2021a; Heuberger and Cohen 2019; Watson et al. 2022). AASs are synthetic derivatives of testosterone. The rationale for use of AASs as performance-enhancing agents is based on their potential to affect an increase in muscle protein synthesis and decrease its breakdown, thereby increasing the skeletal muscle mass and strength gains. Such an effect has been observed when AASs are used in the context of sufficient energy and protein intake combined with resistance training. AAS have not been shown to enhance performance in endurance activities. Numerous oral, injectable, and topical anabolic androgenic products are available in the market. Athletes preferentially use intramuscular injections in supraphysiologic dosages. The effects of AAS on muscle mass and strength gains are seen with higher dosages. Some studies have shown a smaller effect from AAS use in increasing red blood cell mass and oxygen carrying capacity of the blood, improve neuromuscular transmission, and reduce reaction time. Athletes often combine multiple formulations of AAS in a practice called stacking. In a pattern of use called cycling, athletes allow a period of no use. Some athletes

Table 1 Potential health risks associated with AAS use

Category	Health risks
Mental health, behavioral	Aggressiveness, suicidal and homicidal tendencies, depression, hypomania and mania, substance abuse, antisocial behavior, declining cognitive function
Cardiovascular	Lower HDL cholesterol, higher LDL cholesterol, sudden cardiac death, myocardial hypertrophy, fluid retention, and elevated systemic blood pressure
Musculoskeletal	Increased risk for musculotendinous and ligament injuries
Hepatic	Abnormal liver function tests, hepatic dysfunction, tumors, peliosis hepatitis
Specific relevance to females	Hirsutism, clitoromegaly, deep voice, breast atrophy, menstrual irregularity
Specific relevance to children and adolescents	Premature growth plate fusion, shortened ultimate adult height, premature virilization in boys, and masculinization in girls
Specific relevance to males	Decreased sperm count, decreased testosterone level, reduction of testicular size or atrophy, prostatic enlargement
Other	Local infection and abscess from needle use, systemic infections

engage in pyramiding, in which oral and injectable forms are taken in increasingly higher dosages over a period of time. Numerous health risks have been associated with the use of AAS affecting multiple organ systems and overall health status (Table 1).

Peptide Hormones and Their Releasing Factors, Growth Factors and Growth Factor Modulators, Related Substances, Erythropoietin and Agents Affecting Erythropoiesis, and Mimetics

Erythropoietin and agents affecting erythropoiesis (Heuberger and Cohen 2019; Jelkmann 2011; Watson et al. 2022). Erythropoietin (Epo) is a hormone that is predominantly produced in the kidneys, and it is essential for red blood cell production in the bone marrow. A state of tissue hypoxia is a physiologic stimulus for the expression of Epo and stimulation of erythropoiesis. Anemia of chronic kidney disease is a therapeutic indication for the use of recombinant human Epo. Epo and erythropoiesis-stimulating agents (ESAs) increase the red blood cell and hemoglobin mass, which increase VO_2max. The gain in overall aerobic capacity is the basis for the use of Epo and similar agents by athletes participating in endurance activities; however, the overall evidence to support ergogenic effects of Epo and similar erythropoiesis-stimulating agents and methods is weak. Epo and similar agents are administered via subcutaneous route. The oxygen carrying capacity of blood can similarly be increased by blood transfusion or infusion of hemoglobin. In endurance athletes who abuse Epo and other erythropoiesis-stimulating methods, the increased red cell mass when combined with a relative reduction in plasma volume may lead to a state of hyperviscocity, especially in the setting of dehydration, and can result in serious thromboembolic events such as coronary artery occlusions and cerebrovascular accidents.

Growth hormone and similar agents (Chrousos 2021b; Heuberger and Cohen 2019; Seibert and Rao 2018; Watson et al. 2022; Weltman 2011). Growth hormone (GH) is secreted by the anterior pituitary in a pulsatile pattern that varies with age and gender. The anabolic and lipolytic effects of GH are believed to contribute to sports performance enhancement. An increased protein level and a likely reduction in catabolism result in a positive protein balance in skeletal muscles. GH, IGF-I, and insulin act synergistically to affect muscle protein synthesis. When GH is administered in effective dosages to persons with growth hormone deficiency state, several changes are noted that include an increase in muscle mass, muscle strength, maximal exercise performance, and VO_2max. The synergistic action of GH with insulin and IGF-I stimulates muscle protein synthesis. The lipolytic effects of GH are based on several mechanisms – activation

of adenyl cyclase, cAMP-dependent protein kinase, phosphorylation, and hormone-sensitive lipase, as well as enhanced sensitivity of adipocytes to catecholamines. Although administration of GH to adults with GH deficiency has been shown to result in an increase in muscle mass, strength, VO_2max, left ventricular mass, stroke volume and cardiac output, the evidence to support sports performance enhancement, in otherwise healthy athletes, is equivocal. Also, in persons with GH deficiency with a low baseline fitness, the effect of GH administration is relatively magnified. Athletes participating in sports and activities in which muscle size and strength are important contributors to performance use GH in dosages that are up to 4 times those used for medical indications, often in a cycling or stacking pattern. Excessive intake of growth hormone and similar agents is associated with health risks similar to those seen in patients with growth hormone secreting tumors (acromegaly). There is an excessive growth affecting the skeleton, connective tissues, muscles, visceral organs, as well as the skin tissues and glands. Its metabolic effects result in increased glucose production and insulin resistance, impaired glucose tolerance, increased lipolysis, increased tubular resorption of phosphorous and increased urine calcium, and increased salt and water retention.

Chorionic gonadotrophin (CG) and luteinizing hormone (LH) bind to the LHCG receptor, which in males is located in the testis and stimulates testosterone production. There may be increase of testosterone in males when given injection of CG, which may result in effects similar to anabolic agents; however, there is no evidence to support enhancement in sports performance. Adrenocorticotropic hormone (ACTH) is involved in the hypothalamic-pituitary-adrenal axis and is released in response to stress that results in an increased level of cortisol. Free fatty acids are released in response to the increased level of cortisol, resulting in a glycogen sparing effect and potential improvement in endurance performance. No evidence supports such an improvement in sports performance.

Beta-2 agonists (Heuberger and Cohen 2019; Watson et al. 2022). Beta-2 agonists act on the beta-2 receptors in the bronchial and vascular smooth muscle to affect bronchodilation and vasodilation. To a lesser extent beta-2 agonists also act on beta-1 receptors to affect positive ionotropic and chronotropic effects. Beta-2 agonists are available for use in oral and inhalation formulations. There is lack of evidence for any ergogenic effects of inhaled beta-2 agonists. Very high-dose systemic doses of beta-2 agonist agents have been shown to have some effects to increase lean muscle mass, and improved muscle strength and power is short bursts of activity. However, such use is also associated with significant side effects such as muscle tremors and tachycardia that may mitigate performance.

Hormone and metabolic modulators include aromatase inhibitors, anti-estrogenic substances, agents preventing activin receptor IIB activation (Chrousos 2021b; Heuberger and Cohen 2019; Watson et al. 2022). There is very limited evidence-based data on the use patterns and purported ergogenic effects of the agents included in this group of banned agents. Aromatase inhibitors, selective estrogen receptor modulators, and other anti-estrogenic agents all act to reduce circulating estrogen levels and increase testosterone levels, thereby affecting gains in muscle size and strength. The use of myostatin inhibitor agents by athletes is to increase muscle mass by blocking the negative regulatory function of myostatin on muscle growth. The activators of the AMP-activated protein kinase and peroxisome proliferative-activated receptor delta agonists are used for their potential effects on a more effective substrate utilization and energy expenditure, which may result in enhanced performance. Insulin and insulin mimetics are considered metabolic modulators. Exogenous insulin is purported to increase muscle protein synthesis; however, no studies support ergogenic effects of insulin use. Meldonium, which is included under the metabolic modulator category of banned agents, is purported to improve myocardial functional efficiency during exercise. It is believed to act by reducing carnitine, inhibiting mitochondrial beta-oxidation and augmenting the exercise-induced myocardial carbohydrate metabolism.

Diuretics and Masking Agents

Masking agents are not necessarily banned for their purported sports performance enhancement effects, rather for the athlete's use to manipulate the laboratory results for banned agents. For example, diuretics, by dilution of urine, may interfere with laboratory detection of banned agents.

Methods Prohibited at All Times

Manipulation of Blood and Blood Components

Administration of blood and blood components by intravenous injection or infusion is prohibited, both in and out of competition as it will increase red blood cell and hemoglobin mass resulting in enhanced oxygen carrying capacity of the blood (WADA 2023c). Improved oxygen carrying capacity of the blood may have ergogenic effect in endurance activities. In addition, any form of intravascular manipulation of the blood or blood components by physical or chemical means is also prohibited (WADA 2023c).

Chemical and Physical Manipulation

Under this category of prohibited methods include (1) tampering, or attempting to tamper, to alter the integrity and validity of samples collected during doping control and (2) intravenous infusions and/or injections of more than a total of 100 mL per 12-h period except for those legitimately received in the course of hospital treatments, surgical procedures, or clinical diagnostic investigations (WADA 2023c).

Gene and Cell Doping

Under this category of prohibited methods include any methods with a potential to enhance sports performance. These include (1) the use of nucleic acids or nucleic acid analogues that may alter genome sequences and/or gene expression by any mechanism, including but not limited to gene editing, gene silencing, and gene transfer technologies, and (2) the use of normal or genetically modified cells (WADA 2023c). Various physiologic processes with potential to positively affect sports performance could be gene doping manipulation targets (Cantelmo et al. 2020; Cohen-Haguenauer 2022; Lopez et al. 2020). These include (but are not limited to) blood oxygen carrying capacity, improved pain tolerance, enhancement of muscle growth, and musculoskeletal injury repair. Numerous potential gene doping candidate genes associated with sports performance have been identified that target nervous system, vascular system, endocrine system, and hematologic system. Gene doping strategy may be ex vivo (cells from a patient are treated in culture with the therapeutic gene and then administered to patients) or in vivo (the therapeutic gene, in a delivery system, is directly injected into the patient) (Cantelmo et al. 2020; Cohen-Haguenauer 2022; Lopez et al. 2020). Various systems could be targets to affect specific functions. Some examples of putative targets include brain (endorphins to increase pain tolerance), heart (vascular endothelial growth factor (VEGF) to increase cardiac output), liver insulin-like growth factor (IGF-1) and human growth hormone (HGH) to increase energy metabolism, muscles (peroxisome proliferator-activated receptor (PPAR)-myostatin, angiotensin converting enzymes (ACE), actinin alpha 3 (ACTN3) to upgrade muscle quality and vascularization), and bone marrow (erythropoietin and hypoxia inducible factor (HIF) to promote erythropoiesis (Cantelmo et al. 2020; Cohen-Haguenauer 2022; Lopez et al. 2020)).

Substances Prohibited in Competition

Stimulants

Stimulants are used for the treatment of such conditions as attention-deficit hyperactivity disorders, cold symptoms, and allergic reactions. Stimulants are a diverse group of agents that predominantly exert their effects via different neurotransmitter

systems in the central nervous system and to some extent act peripherally on adrenergic receptors (Heuberger and Cohen 2019; Watson et al. 2022). Most stimulants increase the synaptic dopamine and nor-epinephrine levels by inhibition of re-uptake and reduced sequestration. Stimulant action results in enhanced adrenergic response, increased attention and alertness, and increased motor activity. Limited evidence suggests variable degrees of ergogenic effects, in any form when taken orally in sufficiently high doses.

Narcotics

This group of banned agents mainly include opioid analgesics, which are used therapeutically for their strong analgesic effects. Although athletes may use these agents for prevention or reduction of pain from physical activity, these agents are associated with significant side effects that range from nausea, drowsiness, and sedation to respiratory distress and with substantial high doses, coma, and death.

Cannabinoids

Cannabinoids, both natural and synthetic, are prohibited in competition. Its widespread use in general population poses a challenge for its regulation and testing in athletes while participating in sport events. According to the US Centers for Disease Control and Prevention, in the United States, among people aged 12 or older in 2021, 18.7% (or about 52.5 million people) reported using cannabis in the past 12 months. Cannabinoids act on a wide range of neuroreceptors affecting various functions, including movement, sensation, vision, coordination, memory, reward, and judgment (NIDA 2023). This causes the "high" that people feel as well as other effects that include altered senses (for example, seeing brighter colors), altered sense of time, changes in mood, impaired body movement, difficulty with thinking, and problem solving. Given such wide-ranging effects that may impair sports performance, the use of cannabinoids as ergogenic agents is subject to debate. It is likely that its use is based on some degree of calming effect, reduced pain perception, and improved visual perception; however, evidence to support ergogenic effect has not been established.

Glucocorticoids

Glucocorticoids are used in the treatment of such conditions as allergy, anaphylaxis, asthma, and inflammatory bowel disease (Chrousos 2021a). They are available in topical, inhalation, oral, injectable, and rectal use formulations. Injectable, oral, and rectal use of these agents is prohibited in competition. Glucocorticoids have a wide-ranging hormonal and metabolic effects. There is very limited evidence of ergogenic effects of glucocorticoids when taken in injectable, oral, or rectal route. Limited evidence suggests some improvement in specific parameters of muscle strength and time to exhaustion with multiple doses over time (Heuberger and Cohen 2019; Watson et al. 2022). The adrenal cortex is the primary source of endogenous glucocorticoids; cortisol is the principal glucocorticoid hormone secreted by the adrenal cortex. Excessive intake of exogenous glucocorticoids is associated with multiple health risks. These include excess weight gain and obesity, sodium and water retention, immunosuppression, bleeding tendencies, hypocalcemia, osteopenia, deceleration of linear growth in children and adolescents, increased insulin resistance, and increased risk for diabetes (Chrousos 2021a).

Substances Prohibited in Particular Sports

Beta-Blockers

Beta-blocker medications are commonly used in the treatment of high blood pressure and heart failure (Benowitz 2021). Beta-blockers have inhibitory effect on the beta-adrenergic receptors that result in reduced heart rate, reduced anxiety, and reduced motor tremulousness, which may contribute to

improved performance in sports requiring steadiness, a sense of calm, precision, and accuracy (Heuberger and Cohen 2019; Watson et al. 2022). As such, beta-blockers are prohibited in certain sports that include archery, auto sports, billiards, darts, golf, shooting, certain skiing events, and underwater sports (WADA 2023c).

Testing

In addition to laboratory testing of appropriate samples for specific prohibited substances (that include, but not limited to urine, blood or other matrix, including for deoxyribonucleic acid (DNA) or genomic profiling) within the context of sporting event, WADA has developed the Athlete Biological Passport (ABP) to address the challenges of detecting blood doping and steroid doping (WADA 2023d; Krumm et al. 2022). Under the ABP protocol, the athlete undergoes ongoing testing of defined laboratory parameters over an extended period of time based on the detailed standards developed by WADA (WADA 2023c). Timely interpretation of data in ABP allows for a more effective additional targeted testing or investigation. The ABP can be used to complement the analytical methods to further refine and strengthen the anti-doping strategies, for example, testing for erythropoiesis-stimulating agents (ESAs) or homologous blood transfusion. The findings from ABP can be used as a basis for pursuing violation of the anti-doping code. Specific changes in biological markers of doping trending over a long period of time can effectively establish doping by the athlete and hence doping rule violation. ABP comprises a hematological module and a steroidal module (Table 2), (WADA 2023d). The hematological module includes the biological markers of blood doping and helps to establish the prohibited agents or methods used by the athlete to enhance blood oxygen carrying capacity such as ESAs and blood transfusion or manipulation. The steroidal module includes the biological markers of steroid doping and aims to identify use of endogenous anabolic androgenic steroids (EAAS) when administered exogenously and other anabolic agents such as selective androgen receptor modulators (SARMS). The steroid module is also effective in identifying samples, which may have been tempered with or exchanged with the urine of another person.

Table 2 Blood variables included in Athlete Biological Passport

Module	Blood markers
Hematological module	Abnormal blood profile score Hematocrit Hemoglobin Immature reticulocyte fraction Mean corpuscular hemoglobin Mean corpuscular hemoglobin concentration Mean corpuscular volume OFF-hr score Platelets Red blood cell count Red blood cell distribution width (standard deviation) Reticulocyte count Reticulocyte percentage White blood cells
Steroidal module	Androsterone Etiocholanolone 5α-androstane-3α, 17β-diol 5β-androstane-3α, 17β-diol Testosterone Epitestosterone Following ratios: Testosterone/Etiocholanolone Androsterone/testosterone Androsterone/Etiocholanolone 5αadiol/5βadiol 5αadiol/epitestosterone

Adverse Analytical Findings (AAFs) are findings on laboratory analysis that deviate from established norms and are considered indicative of doping. Table 3 provides a summary of substances identified as AAFs in each drug class (WADA 2021).

Note: The AAFs in this report are not to be confused with adjudicated or sanctioned anti-doping rule violations, as the data reported herein may contain findings that underwent the Therapeutic Use Exemption approval process or multiple findings on the same athlete.

The Executive Summary of the 2021 report highlights the following findings and trends:

- A 61.2% increase in the number of samples (both urine and non-ABP blood samples) analyzed and reported into Anti-doping Administration and

Table 3 WADA reported adverse analytical finding: 2021 report

Substance group or method	Occurrences	Percentage of reported findings
Anabolic agents	875	40
Stimulants	348	16
Diuretics and other masking agents	310	14
Hormone and metabolic modulators	247	11
Glucocorticoids	124	6
Beta-2 agonists	68	3
Cannabinoids	92	4
Peptide hormones, growth factors, and related substances	98	4
Narcotics	22	1
Beta-blockers	9	0.4
Enhancement of oxygen transfer	4	0.2
Chemical and physical manipulation	0	0

Management System (ADAMS): 149,758 in 2020 to 241,430 in 2021. It should be noted that the COVID-19 pandemic significantly impacted the number of samples collected in 2020.

- An increase in the total number of samples analyzed and reported by nearly all WADA-accredited laboratories and WADA-approved laboratories into ADAMS in 2021 compared to 2020.
- An increase in the total number and percentage of non-ABP blood samples analyzed: 7.3% (10,940 of 149,758 samples) in 2020 to 9.3% (22,398 Blood and dried blood spot (DBS) samples out of 241,430).
- An increase of 36% in the number of ABP blood samples analyzed: 22,666 in 2020 to 30,821 in 2021.
- A decrease in the total percentage of AAFs: 0.67% in 2020 (1009 AAFs from 149,758 samples) to 0.65% (1560 AAFs from 241,430 samples).
- An increase in the total number of AAFs for growth hormone (GH): 1 in 2020 to 7 in 2021, including the first reported AAF for a GH biomarker.

Nutrition for Performance and Recovery

The International Olympic Committee (IOC), Academy of Nutrition and Dieticians, Dieticians of Canada, American College of Sports Medicine (ACSM), Australian Institute of Sports (AIS), International Society for Sport Nutrition (ISSN), and National Institutes of Health Office of Dietary Supplements (NID ODS) provide guidance on nutrition in general and use of nutritional or dietary supplements in particular, for athletic performance and recovery, key findings and recommendations from which are summarized in the following sections (ACSM 2016; Kirksick et al. 2018; Maughan et al. 2018; Papadopoulou 2020; AIS 2023; NIH 2023a, b, c, d).

Macronutrients

Energy Needs and Carbohydrates

Sufficient and regular carbohydrate intake is essential to meet the daily basal as well as the increased energy need associated with exercise. Because of limited capacity for the body to store carbohydrates, it is possible to modulate body carbohydrate reserves by varying intake and expenditure from exercise. Since carbohydrates can be utilized by both the anaerobic and oxidative pathways, carbohydrates provide key substrate for muscular work for a wide range of exercise intensities. In addition to their role in muscular activity, carbohydrates serve as the key fuel for brain activity. Carbohydrates provide a relatively higher yield of adenosine triphosphate per volume of oxygen delivered to the mitochondria when compared with fat, which allows for a greater exercise efficiency (ACSM 2016). For a high level of performance during sustained or intermittent high-intensity exercise, sufficient level of both glycogen stores and blood glucose must be available. Such energy needs can be met most effectively by ensuring sufficient intake of carbohydrates before, during, and following exercise.

Daily needs for carbohydrates needed for fuel and recovery vary depending on the intensity of

the activity and range from 3–5 g/kg of athlete's body weight per day for light (low intensity or skill-based activities) to 8–12 g/kg/day for very high (generally 4–5 h or more/day of moderate-to-high intensity activities). For optimal availability of carbohydrates to meet the energy needs, the timing of intake may be before or during the session or during the recovery period following a session. The practice of periodic intake of small amounts of carbohydrates orally during exercise (mouth sensing) has been shown to enhance the effectiveness of the workout and pacing strategies by its actions on the central nervous system. Low fiber or residue and low glycemic index carbohydrates are recommended. For optimal performance in competition or during high-quality and/or high-intensity exercise sessions, the ACSM position statement provides more specific guidance for the amount of carbohydrate targets needed for acute fueling strategies (Table 4).

Some of the key elements of guidance provided by the International Society of Sports Nutrition (ISSN) include the following considerations (Kirksick et al. 2018):

- It takes approximately 4 h for the ingested carbohydrates to be fully assimilated to restore muscle and liver glycogen stores.
- Pre-event carbohydrate meal is best consumed about 4–6 h prior to the event.
- For rapid recovery and replenishing muscle and liver glycogen following exercise session or sporting event ISSN recommends two strategies:
  (1) consumption of 1.2 g of high glycemic index carbohydrate/kg of body weight per hour post event,
  (2) consumption of a combination of 0.8 g/kg/h of carbohydrate and 0.2–0.4 g/kg/h of protein and adding 3–8 mg/kg of caffeine.
- Athletes who engage in high-intensity events lasting more than 90 min may oxidize carbohydrates at a rate of 1–1.1 g per minute. In such scenarios, the ISSN statement recommends consumption of carbohydrates at a rate of 30–60 g per hour in the form of a solution that contains 6–8% carbohydrate, 10–35 mmol/L sodium, and 3–5 mmol/L potassium, which is typically accomplished by ingesting 6–12 fluid ounces of the solution every 10–15 min throughout the exercise session.

Table 4 ACSM guidelines for carbohydrate intake

Acute fueling strategies	Carbohydrate targets
General fueling up for preparation for events <90 min of exercise	7–12 g/kg per 24 h as for daily fuel needs
Carbohydrate loading for preparation for events >90 min of sustained or intermittent exercise	36–48 h of 10–12 g/kg body weight per 24 h preceding the event
Speedy refueling <8 h recovery between 2 fuel demanding sessions	1–1.2 g/kg/h for first 4 h then resume daily fuel needs
Pre-event fueling before exercise lasting >60 min	1–4 g/kg consumed 1–4 h before exercise
During brief exercise lasting <45 min	No specific need
During sustained high-intensity exercise lasting 45–75 min	Small amounts including mouth rinse
During endurance exercise lasting 60–150 min, including 'stop and start' sports	30–60 g/h
During ultra-endurance exercise lasting >150–180 min	Up to 90 g/h

Protein

Appropriately timed (relative to the time of an exercise session) intake of sufficient amount of protein stimulates muscle protein and metabolic protein synthesis and also affect favorable structural adaptations of connective tissue and bone. The primary underlying basis for protein synthesis and adaptive changes is a rise in blood leucine concentration concomitant with intake of amino acids needed to synthesize new proteins. Following a high-quality session of resistance exercises, there is an upregulation of muscle protein synthesis lasting for subsequent 24 h with an increased responsiveness to dietary protein intake. Similar response following aerobic or other types of

exercise stimulate synthesis of different types of proteins.

A dietary protein intake of 1.2–2 g/kg/day is recommended for optimal favorable effect on the metabolic adaptation, repair, and remodeling and allows for protein turnover. In a setting of short-duration very high-intensity training or when overall energy intake is low, much higher doses of protein intake may be of benefit to induce favorable response.

Concomitant intake of sufficient amount of carbohydrates is essential to spare amino acids for protein synthesis. The consumption of adequate energy, particularly from carbohydrates, to match energy expenditure, is important so that amino acids are spared for protein synthesis and not oxidized. Muscle protein synthesis is optimized with intake of 0.25–0.30 g/kg/body weight of high biological value protein within 2 h post-exercise. Further adaptation to training is seen with consumption of similar amounts of protein every 3–5 h following exercise.

Consumption of milk-based proteins after resistance exercise has been shown to be highly effective in affecting muscle protein synthesis, improving muscle strength and lean mass. The superior effect of dairy proteins is primarily attributed to its leucine content as well as digestion and absorptive kinetics of branched-chain amino acids in fluid-based dairy products. Other sources of protein include lean meat and other animal protein as well as a wide variety of protein supplements that contain variable amounts of whey, casein, soy, or egg proteins.

Fat

Athletes should be advised not to substantially restrict fat intake. Fat is essential component of a well-balanced diet that provides fuel for energy and facilitates the absorption of fat-soluble vitamins. Fat also provides the essential elements needed for proper development of cell membranes. Plasma-free fatty acids, adipose tissue, and muscle triglycerides serve as substantial sources of energy for exercise, especially for endurance training.

Fluids and Electrolytes

Sufficient and timely intake of fluid and electrolytes is essential for optimal exercise performance and recovery as well as to prevent dehydration and heat-related spectrum of illnesses in hot ambient conditions. The observed effects of fluid deficits depend on the amount of loss and the environmental contexts (Table 5) (ACSM 2016).

For short-duration activity, generally less than 60 min of low to moderate intensity, water is an adequate source of fluid replacement. For higher intensity and longer duration activities, energy needs and electrolyte loss should be considered as a component of maintaining fluid and electrolyte balance and replacement strategy. Flavored drinks also enhance palatability and adherence to fluid intake recommendations. As a general guideline, a solution/sports beverage should contain 6–8% carbohydrates, 10–35 mmol/L of sodium, and 3–5 mmol/L of potassium (Kirksick et al. 2018). It is important to note that multiple factors affect fluid loss during physical activity and include the type of exercise, intensity and duration of exercise, altitude, environmental heat and humidity, baseline fitness of the athlete, acclimatization status, and sweat rate. The American College of Sports Medicine position statement

Table 5 Effects of fluid deficits

Fluid deficit as a percentage of body weight	Observed effects
2%	Compromised cognitive function, deterioration of aerobic exercise performance, especially in hot weather
3–5%	Deterioration of anaerobic exercise performance, deterioration in high-intensity activities and sport-specific skills, low aerobic exercise performance
6–10%	Severe dehydration, decreased cardiac output, decreased sweat production, reduced peripheral blood flow, increased risk for heat-related illnesses in hot environment

recommends the following to guide fluid and electrolyte intake (ACSM 2016):

- Fluid intake of 5–10 ml/kg body weight in the 2–4 h preceding the exercise is recommended.
- Fluid and electrolyte replacement during exercise should match to replace losses, especially additional losses in sweat. As the sweat rates vary from 0.3 to 2.4 L/h during exercise depending up on multiple factors, fluid intake often needs to be tailored to individual need and circumstance.
- For most range of exercise activities, a loss of 1 kg of body weight generally indicates approximately 1 L of sweat loss.
- The overall goal is to prevent fluid loss not to exceed 2% of body weight. As a general guideline, consumption of 0.4–0.8 L/h of fluid-electrolyte solution would be a good basis for most athletes.
- Fluid and electrolyte losses continue for a variable duration post-exercise and require continued replacement. To replace such a fluid loss, athletes should consume approximately 1.25–1.5 L fluid for every 1 kg BW post-exercise.

Micronutrients

Vitamin D and Calcium

Adequate daily intake and maintaining sufficient levels of vitamin D and calcium are essential to the growth, development, and overall health of the musculoskeletal system (ACSM 2016; Kirksick et al. 2018; NIH 2023a, b; Rawson et al. 2018). Vitamin D enhances absorption of calcium in the gut, and along with calcium, it is essential in the acquisition of optimal bone mineral density. Adequate intake of vitamin D and calcium is especially important in children and adolescents as most bone mineral density (BMD) is acquired during the second decade of life. In addition to its role in bone health, calcium is essential for optimal neuromuscular functioning and normal blood clotting. Although there is not a consensus on the optimal blood level of vitamin D (as measured in the form of 25-hydroxy-vitamin D or 25(OH)D), most guidelines suggest that a level less than 75 nmol/L indicates vitamin insufficiency and a level less than 50 nmol/L indicates vitamin D deficiency. Athletes with vitamin D insufficiency or deficiency should be evaluated for factors that might have contributed to low vitamin D levels, and these factors should be addressed including dietary habits. Vitamin D supplements sufficient to correct the deficiency are recommended, the dose for which will depend on the level of deficiency. General recommendations for dietary intake of vitamin D for otherwise healthy adults range from 800 IU to 2000 IU, and for calcium, the recommendation is 1000 mg/d for men and 1200 mg/d for women. Although it is important to maintain adequate vitamin D and calcium intake for optimal bone and muscle health, there is insufficient evidence to support routine supplementation for any ergogenic effects.

Iron

Iron is an essential mineral with key roles in the hematologic, metabolic, respiratory, and immune functions (ACSM 2016; Kirksick et al. 2018; NIH 2023c; Rawson et al. 2018). Of specific relevance in athletes, iron being a component of hemoglobin in red blood cells is necessary for optimal oxygen carrying capacity of the blood. Iron deficiency with or without anemia (low iron stores without anemia) results in an impaired oxygen carrying capacity and adversely affects exercise performance, contributing to poor training adaptations. Iron deficiency in athletes may result from poor dietary habits, low biologic availability of iron, increased need during periods of rapid growth, menstrual blood loss, training at high altitudes, foot strike hemolysis, and increased losses of iron in sweat, urine, or feces. A serum ferritin level of <20 micrograms/L is generally considered indicative of iron deficiency. Athletes with iron deficiency should be evaluated clinically, and any contributing factors should be identified and addressed as part of overall management. Iron deficiency should be corrected by appropriate iron supplementation. Although correcting iron

deficiency would improve overall exercise capacity and improve performance, there is no evidence to support ergogenic role of iron in athletes without any iron deficiency. The general recommendation for daily dietary intake of iron is 8 mg for men and 18 mg for women.

Nutritional Supplements

The prevalence of nutritional supplement use for sports performance enhancement by athletes varies depending upon multiple factors. The exact prevalence is difficult to ascertain; however, it is believed to be widespread and tends to be higher among high-level elite athletes. A large number of nutritional supplements are marketed with numerous claims for their efficacy in improving sports performance. The International Olympic Committee, Academy of Nutrition and Dietetics, Dieticians of Canada, American College of Sports Medicine, Australian Institute of Sports, and the US Department of Health and Human Services National Institutes of Health Office of Dietary Supplements (USDDHS NIH ODS) provide research reviews and guidelines for specific supplements used by athletes (ACSM 2016; AIS 2023; Kirksick et al. 2018; Maughan et al. 2018; NIH 2023d; Rawson et al. 2018; Smith-Ryan et al. 2020). From a very large number of supplements being marketed, only a handful have been shown to affect improvement on specific parameters relevant to sports performance. These include caffeine, creatine, nitrate, beta-alanine, and sodium bicarbonate.

Caffeine

Caffeine is a methylated xanthine naturally found in coffee, tea, cacao pods (the source of chocolate), and other herbal/botanical sources, such as guarana, kola (or cola) nut, and yerba mate. Caffeine is primarily a central nervous system stimulant, which also has some stimulatory effects on heart and muscles. Caffeine acts by binding to adenosine receptors on cells and thereby blocking the action of neuromodulator adenosine, which has a sedative effect. This results in the person experiencing heightened alertness, mitigation of fatigue, and a reduced perception of pain and exertion. During the early stages of endurance exercise, caffeine might mobilize free fatty acids as a source of energy and spare muscle glycogen.

Caffeine is the primary ingredient in energy drinks and numerous other products marketed for their ergogenic effects. Pure powdered caffeine is available as a dietary supplement and is very potent. A single tablespoon contains 10 g caffeine, and an acute oral dose of 10–14 g caffeine (approximately 150–200 mg/kg) can be fatal. Caffeine, when taken in a dose of about 2–6 mg/kg body weight prior to exercise, has been shown to improve endurance as well as strength and power in high-intensity team sports. For potential ergogenic effect, caffeine should be consumed about 15–60 min prior to exercise. Caffeinated drinks consumed during endurance activity of long duration have been shown to improve performance. A generally accepted recommendation for caffeine intake by otherwise healthy adults is about 400–500 mg/day. Higher doses of caffeine (500 mg/day or more) do not necessarily enhance performance and result in adverse effects including disturbed sleep, irritability, headache, anxiety, restlessness, nausea, vomiting, tachycardia, and dysrhythmias. Caffeine does not induce diuresis or increase sweat loss during exercise and therefore does not reduce fluid balance in the body that would adversely affect performance. Caffeine is rapidly absorbed from buccal membranes and gastrointestinal tract, is distributed in the brain and rest of the body, reaching a peak concentration in the blood within 45 min of ingestion, and has a half-life of 4–5 h.

The International Olympic Committee restricts the use of caffeine up to an amount that would not result in a urine concentration that exceeds 12 mcg/ml of urine. The National Collegiate Athletic Association prohibits use of caffeine from any source in amounts that would lead to urine concentrations exceeding 15 mcg/ml. (Consuming about 500 mg caffeine produces a urinary caffeine concentration of 15 mcg/ml within 2–3 h.)

Creatine

Creatine is produced endogenously and to a lesser extent obtained from animal-based food such as beef, pork, and salmon. The liver and kidneys synthesize about 1 g/day creatine from the amino acids, glycine, arginine, and methionine. The primary function of creatine is to generate adenosine triphosphate (ATP) and thereby supply the muscles with energy, particularly for short-term events. The ergogenic effects of creatine are derived primarily from enhanced skeletal muscle function. Creatine has been shown to improve muscle performance by increasing stores of the phosphocreatine used to generate ATP at the beginning of intense exercise; accelerating the re-synthesis of phosphocreatine after exercise; reducing the degradation of adenine nucleotides and the accumulation of lactate; and/or enhancing glycogen storage in skeletal muscles. For ergogenic effects in short-duration, high-intensity events, as well as to allow for long-term adaptations to structured workloads over time, creatine needs to be supplemented in the diet in higher amounts for a longer period. Creatine enhances performance during repeated short bursts of high-intensity, intermittent activity that predominantly utilize the anaerobic pathway in which energy is derived from the ATP-creatine phosphate energy system.

A typical protocol for creatine supplementation in adults, regardless of sex or body size, consists of a loading phase for 5–7 days, when 20 g/day creatine monohydrate is consumed in four portions of 5 g each, followed by a maintenance phase of 3–5 g/day. Another creatine supplementation protocol consists of taking single doses of about 3–6 g/day (0.03–0.1 g/kg body weight) for 3–4 weeks, without a loading phase, to produce ergogenic effects. Creatine is metabolized in the body to creatinine and eliminated by kidneys. Creatine monohydrate, which is 88% creatine by weight, is the most widely used and studied form. Creatine used based on generally practiced protocols has not been associated with any significant health risks. Some athletes may experience bloating, water retention, weight gain, and gastrointestinal symptoms. Some studies have reported that creatine supplementation may cause muscle stiffness and decreased musculotendinous flexibility in some athletes. Maintaining sufficient fluid intake and hydration have been shown to mitigate short-term side effects associated with creatine.

Nitrate

Beets are a rich source of inorganic nitrate. Many products of beetroot are available in the form of juice or juice concentrate and powdered form. Ingested nitrate exerts its ergogenic effects when it is converted in the body to nitric oxide. Nitric acid is a potent vasodilator that can increase blood flow and the delivery of oxygen and nutrients to the exercising skeletal muscle, reducing the oxygen cost of submaximal exercise, attenuating the adenosine triphosphate (ATP)-creatine phosphate energy system's cost associated with skeletal muscle force production, and improving oxidative phosphorylation in mitochondria (Jaffrey 2021). Most studies have used 500 ml/day (about 2 cups) of beetroot juice taken once (about 2.5–3 h before exercise) or daily for up to 15 days. This amount of juice provides about 5–11 mmol (or 310–682 mg) nitrate, depending on the product. Potential benefits persist for up to 24 h after ingestion. Consumption of moderate amounts of beetroot juice daily has not been shown to be unsafe or associated with health risks.

Beta-Alanine

Beta-alanine, an amino acid, is the rate-limiting precursor to the synthesis of carnosine skeletal muscles. Carnosine helps buffer changes in muscle pH from the anaerobic glycolysis that provides energy during high-intensity short-duration exercise. However, this metabolic process results in the buildup of hydrogen ions as lactic acid accumulates and dissociates to form lactate,

leading to reduced force and to fatigue. A higher concentration of muscle carnosine leads to greater potential attenuation of exercise-induced decrease in pH.

Beta-alanine is produced in the liver, and relatively small amounts are present in animal-based foods such as meat, poultry, and fish. Estimated dietary intakes range from none in vegans to about 1 g/day in heavy meat eaters. Although carnosine is present in animal-based foods, oral consumption of carnosine is an inefficient method of increasing muscle carnosine concentrations because it is digested into its constituent amino acids. Consumption of beta-alanine supplement is needed to increase the amount of carnosine in the body.

The evidence supporting ergogenic effects of beta-alanine is equivocal. Some studies suggest that beta-alanine consumption could provide small performance benefits in competitive events requiring high-intensity effort over a short period, such as rowing, swimming, and team sports that involve repeated sprints and intermittent activity, while other studies have found no such benefits. Evidence is also equivocal for any ergogenic effects of beta-alanine in endurance activities.

Beta-alanine supplement in 4–6 g/day consumed for 2–4 weeks has been shown to improve high-intensity exercise performance that lasts over 60 s, especially in time-to-exhaustion tasks; however, performance improvement is at best minimal in exercise tests lasting more than 4 min, which primarily utilize aerobic metabolic pathways to meet energy demands. Ingestion of beta-alanine at 4–6 g/day for 4 weeks has been shown to increase muscle carnosine level by about 40%; however, the evidence is inconsistent for a clear relationship between the dose of beta-alanine and magnitude of performance improvement. Although beta-alanine is considered generally safe, ingestion exceeding 10 mg/kg body weight has been reported to be associated with moderate to severe paresthesia, especially tingling, prickling, or burning sensation in the face, neck, back of the hands, and upper trunk lasting 60–90 min. Beta-alanine supplement is best taken with meals to augment muscle carnosine levels, and in divided lower doses or as sustained-release form, if paresthesia occurs.

Sodium Bicarbonate

The consumption of sodium bicarbonate (commonly known as baking soda) transiently raises the blood pH by acting as a buffering agent, which leads to metabolic alkalosis. The mechanism by which induced alkalosis leads to an ergogenic response to exercise is unclear. During short-duration, high-intensity exercise, when muscles derive energy predominantly in the form of ATP via anaerobic glycolysis, there is accumulation of hydrogen ions and resultant metabolic acidosis. Metabolic acidosis contributes to muscle fatigue. Oral supplementation with sodium bicarbonate mitigates the development of metabolic acidosis which contributes to muscle fatigue and improves exercise performance in short-duration, high-intensity events. Although an optimal dose of sodium bicarbonate supplement for its ergogenic effects is not clear, a generally accepted practice is to take 300 mg/kg body weight (4–5 teaspoonfuls of baking soda), 1–2 h before exercise, in a one time or divided doses. Although sodium bicarbonate is well tolerated, some athletes experience nausea, abdominal pain, diarrhea, and vomiting. Sodium bicarbonate is 27.4% sodium by weight, and a typical dose of 300 mg/kg body weight (for average 70 kg person) would contain approximately 5750 mg sodium. Consumption of such a large amount of sodium with fluid would result in a state of short-term hyperhydration. Such a temporary hyperhydration could be useful in activities where large sweat losses might otherwise lead to significant fluid deficits; however, the slight increase in body weight from fluid retention might hinder performance in other sports. The safety and effectiveness of long-term use of sodium bicarbonate as an ergogenic aid over months or longer is poorly understood.

Conclusion

Athletes use a number of drugs and nutritional supplements for their ergogenic effects. Various sports governing bodies and professional scientific societies have published guidelines for optimal nutrition for athletes as well as reviewing the claims of ergogenic effects of a large number of nutritional supplements available in the market. The WADA through its anti-doping program sets forth specific anti-doping rules and principles that are to be followed by organizations responsible for adopting, implementing, or enforcing anti-doping rules within their authority. Although it is important to maintain adequate vitamin D and calcium intake for optimal bone and muscle health, there is insufficient evidence to support routine supplementation for any ergogenic effects. Similarly, any iron deficiency should be evaluated and treated; however, iron supplementation is not recommended for any ergogenic role. From a very large number of supplements being marketed, only a handful have been shown to affect improvement on specific parameters relevant to sports performance. These include caffeine, creatine, nitrate, beta-alanine, and sodium bicarbonate.

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